

2025-11-01

EP001: Physical System Description

Double-Inverted Pendulum Control Theory

Double-Inverted Pendulum Learning Module

March 30, 2026

section0 Introduction

subsection0.0 Overview

The double inverted pendulum is a classic benchmark in nonlinear control.

- It is unstable in open loop.
- It has coupled rotational dynamics.
- It has one actuator and two rotational states.

subsection0.0 System Characteristics

The plant is nonlinear and underactuated:

enumiNon-minimum phase behavior can appear near upright equilibrium.

0. enumiFull state feedback improves achievable robustness margins.
0. enumiDisturbances strongly affect transient response.

section0 Physical System Description

subsection0.0 System Configuration

The model consists of a cart, two serial rigid links, and one control force.

0. Cart mass: m_c

- Link 1 mass: m_1 , length: l_1 , angle: θ_1
- Link 2 mass: m_2 , length: l_2 , angle: θ_2

subsection0.0 Physical Constraints

enumiCart travel: $x \in [x_{\min}, x_{\max}]$

0. enumiForce bounds: $u \in [u_{\min}, u_{\max}]$
0. enumiJoint limits near safe mechanical stops if present

section0 Mathematical Model

subsection0.0 State Vector Definition

State vector choice is:

$$\text{equation} x = [q, \dot{q}]^T = [x, \theta_1, \theta_2, \dot{x}, \dot{\theta}_1, \dot{\theta}_2]^T. \quad (0)$$

subsection0.0 Equations of Motion

A compact form is used:

$$\text{equation} M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = Bu + d. \quad (0)$$

definition The equation is valid under rigid-link and rigid-joint assumptions.

recordtarget subsection0.0 Inertia Matrix

The inertia matrix $M(q)$ is symmetric and positive definite.

$$\text{equation} M(q) = \begin{bmatrix} M_{11} & M_{12} & M_{13} \\ M_{12} & M_{22} & M_{23} \\ M_{13} & M_{23} & M_{33} \end{bmatrix}. \quad (0)$$

subsection0.0 Coriolis and Centrifugal Matrix

Coriolis terms are configuration and velocity dependent.

$$\text{equation} C(q, \dot{q})\dot{q} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}. \quad (0)$$

section0 Nonlinearity Analysis

subsection0.0 Configuration-Dependent Inertia

The inertia changes with configuration and couples all coordinates.

- 0. enumiDependence on $\cos \theta_1$, $\cos \theta_2$, and $\cos(\theta_1 - \theta_2)$
- 0. enumiDeterminant and condition number vary with q
- 0. enumiIll-conditioning near singular alignment may occur

subsection0.0 Trigonometric Nonlinearity

theorem Gravity and coupling create trigonometrical nonlinearities.

recordtarget The dominant terms include $\sin \theta_i$ and $\cos \theta_i$.

section0 System Parameters

subsection0.0 Nominal Parameter Values

subsection0.0 Parameter Uncertainty

- 0. Mass and friction values vary with payload.
- Length tolerances affect effective lever arms.
- Sensing noise biases derivative terms.

table

Table 0: Nominal DIP System Parameters

Parameter	Value
m_c	1.0 kg
m_1	0.5 kg
m_2	0.3 kg
l_1	0.4 m
l_2	0.3 m
g	9.81 m/s ²

section0 Linearized Model

subsection0.0 Equilibrium Points

Equilibrium near upright configuration is commonly chosen:

$$equation x_e = [0, 0, 0, 0, 0]^T, \quad (0)$$

with other valid equilibria depending on control objective.

subsection0.0 Linearized State-Space Representation

After small-angle approximation the local model is:

$$equation \dot{x} = Ax + Bu + Ew. \quad (0)$$

- Stability of A determines local behavior.
- Controller gains depend on performance versus robustness trade-offs.

section0 Numerical Properties

subsection0.0 State Scaling

enumiAngles are typically in radians.

0. enumiVelocities are in rad/s.
0. enumiScaling improves optimization conditioning.

subsection0.0 Time Scaling

0. enumiChoose step size based on highest mode frequency.
0. enumiEnsure numerical integrator accuracy near impact events.
0. enumiUse fixed-step simulation for reproducibility.

section0 Summary and Key Takeaways

subsection0.0 System Overview

The double inverted pendulum is a stiff, coupled, underactuated plant.

subsection0.0 Key Challenges

- 0. enumiNonlinearity dominates wide operating ranges.
- 0. enumiStrong coupling changes control authority with configuration.
- 0. enumiUnmodeled disturbances degrade near-boundary stability.

subsection0.0 Control Implications

theoremControl design requires balanced trade-offs between competing objectives.

- recordtargetLinear designs simplify synthesis.
- Sliding and adaptive methods improve robustness.
- Constraint-aware tuning is mandatory in hardware loops.